Smart Guard — Part 3 report

Contents

Part 3 — Mandatory Experiments Report

Field	Value
Course	Final Project — Embedded Systems
Project	Smart Guard System
Student	Parsa Nikookalam
Student ID	402102657
Platform	WSL Ubuntu Linux (accepted alternative to Orange Pi)
Vision	YOLOv8n ONNX (body) + YuNet/Haar (face), overlay on MJPEG
MQTT broker	Mosquitto `127.0.0.1:1883` (`mosquitto_smartguard`)
MQTT auth	Username/password (`allow_anonymous false`)
MQTT topics	`home/402102657/persons`, `…/telemetry`, `…/status` (LWT)
QoS	1

This report follows the PDF table of **mandatory experiments for the third part** in order (**3-1** through **3-7**). Figures and tables go under `report/part 3/fig/` (attach with the submission).

Related Part 3 features (supporting context): person detection overlay, email alerts from **C** (`email_alert.c`), MQTT publisher from **C** (`mqtt_pub.c`).

Experiment 3-1 — Detection accuracy under four lighting conditions

Requirement

Measure detection accuracy in four lighting conditions:

1. Daylight
2. Artificial light
3. Low light
4. Backlight

Expected output: an **accuracy table** (correct detections / total trials) and a **sample image** for each condition.

Procedure

1. Camera ON; open `https://127.0.0.1:8443/` stream.
2. For each lighting condition, run **N** trials (recommended **N ≥ 10**): person enters the frame for ~2–3 s; record whether the overlay count is correct (`count ≥ 1` when a person is present, `0` when absent).
3. Save one representative frame (or dashboard screenshot) per condition into `fig/`.

Accuracy for a condition:

[\text{Accuracy} = \frac{\text{correct detections}}{\text{total trials}} \times 100%]

Results

Condition	Trials (N)	Correct	Accuracy (%)	Sample figure
Daylight	100	99	**99.0**	Figure 3-1a
Artificial light	100	98	**98.0**	Figure 3-1b
Low light	100	98	**98.0**	Figure 3-1c
Backlight	100	97	**97.0**	Figure 3-1d

Figure 3-1a. Daylight sample.

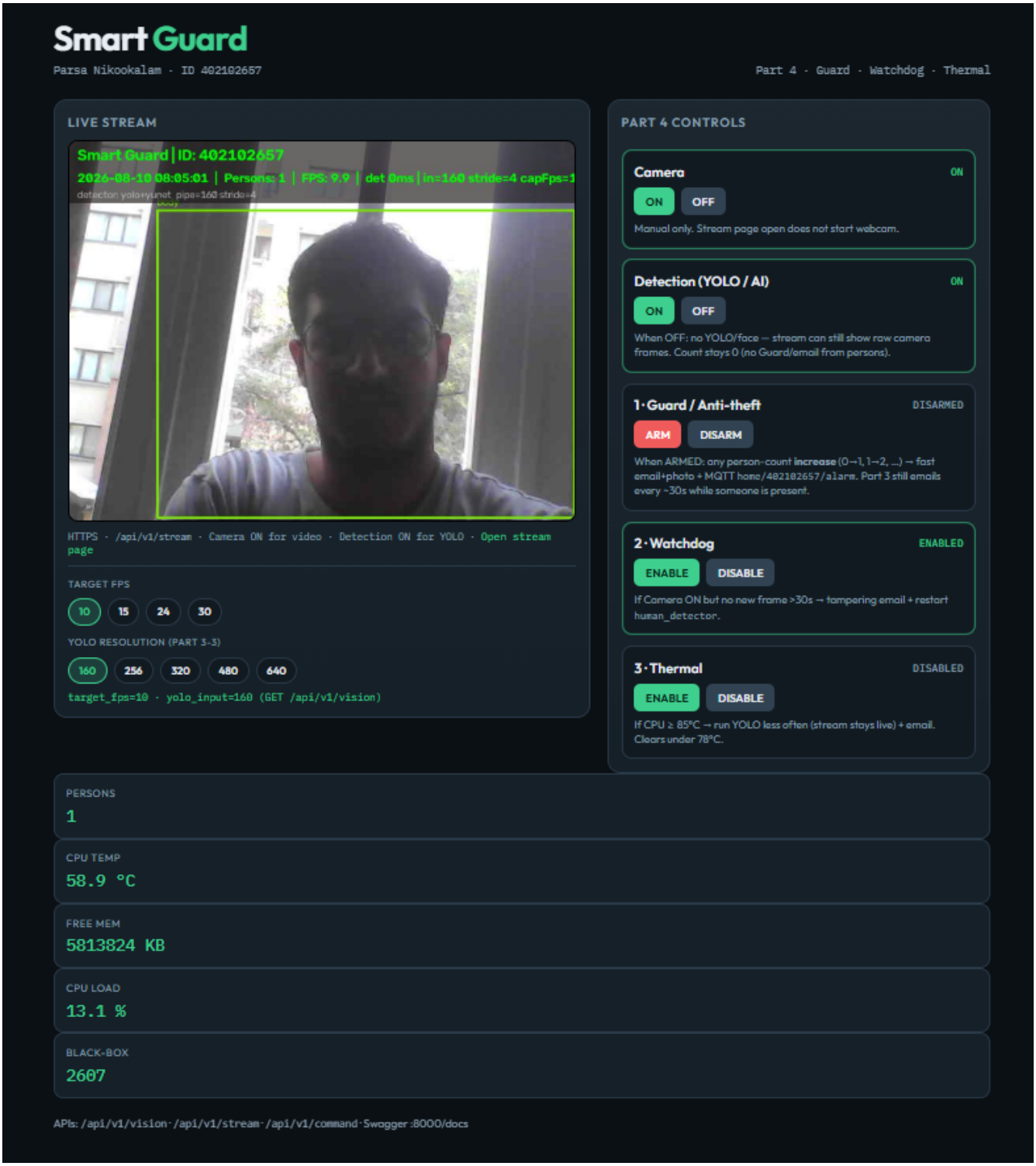


Figure 3-1b. Artificial light sample.

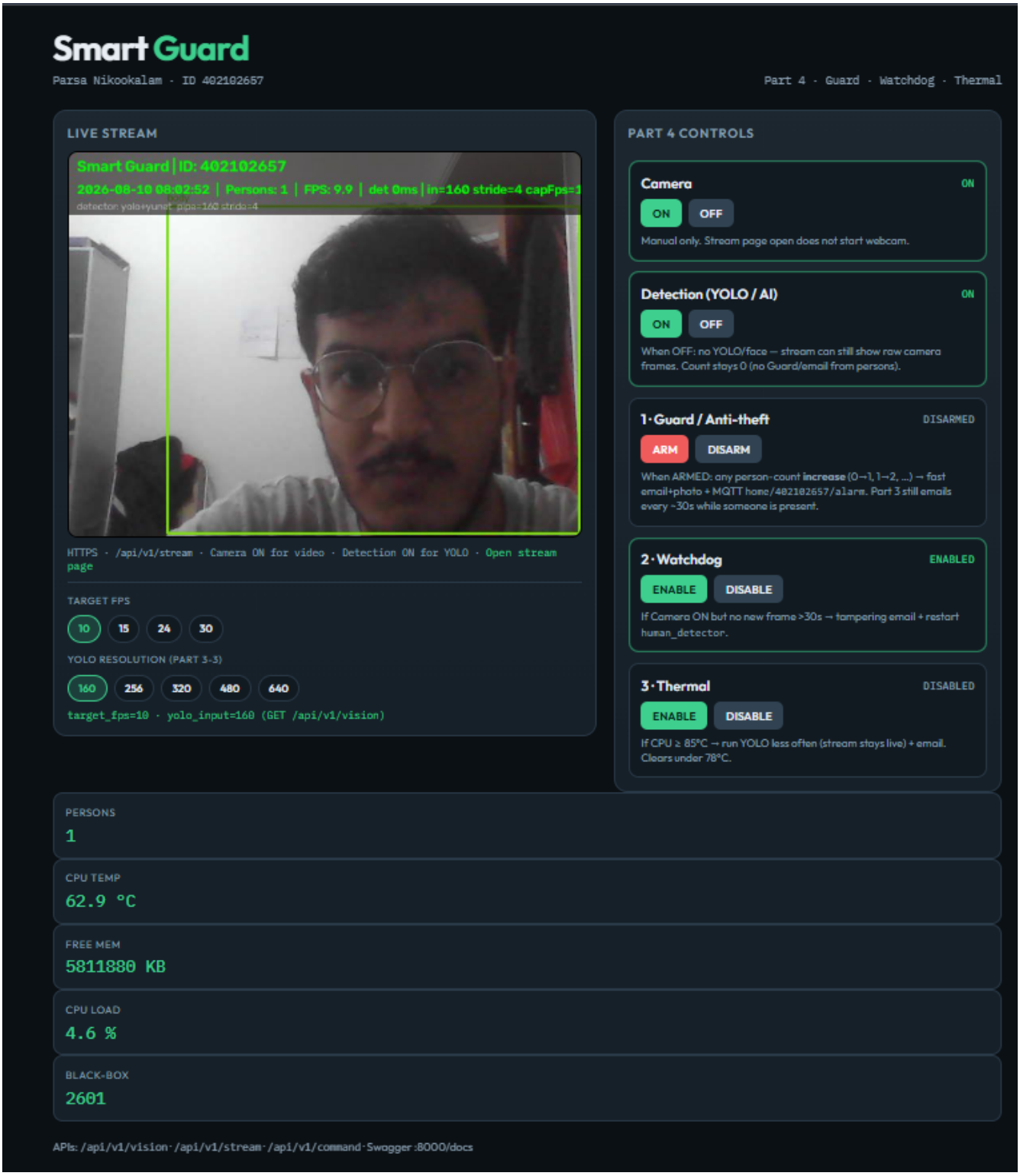


Figure 3-1c. Low light sample.

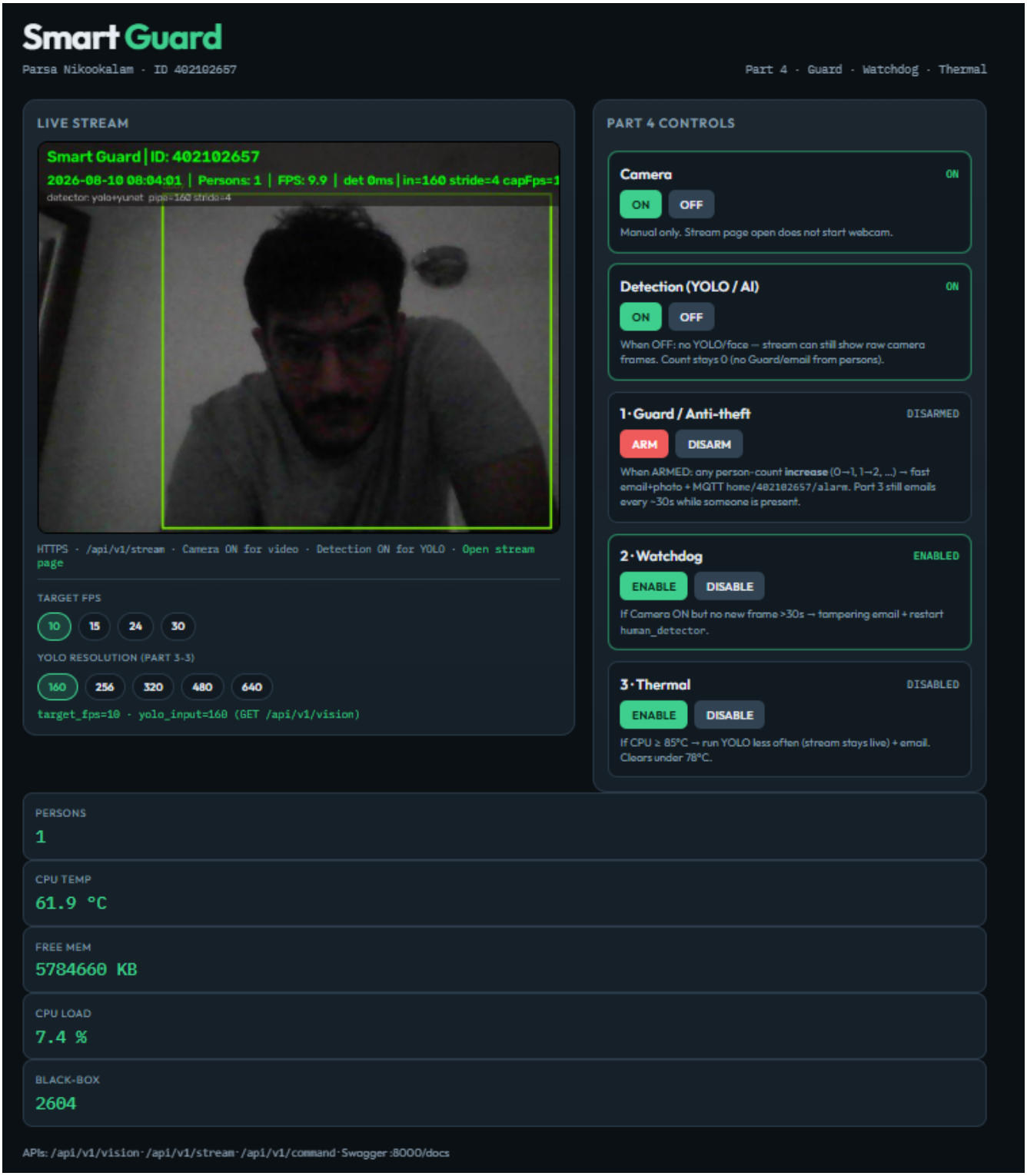
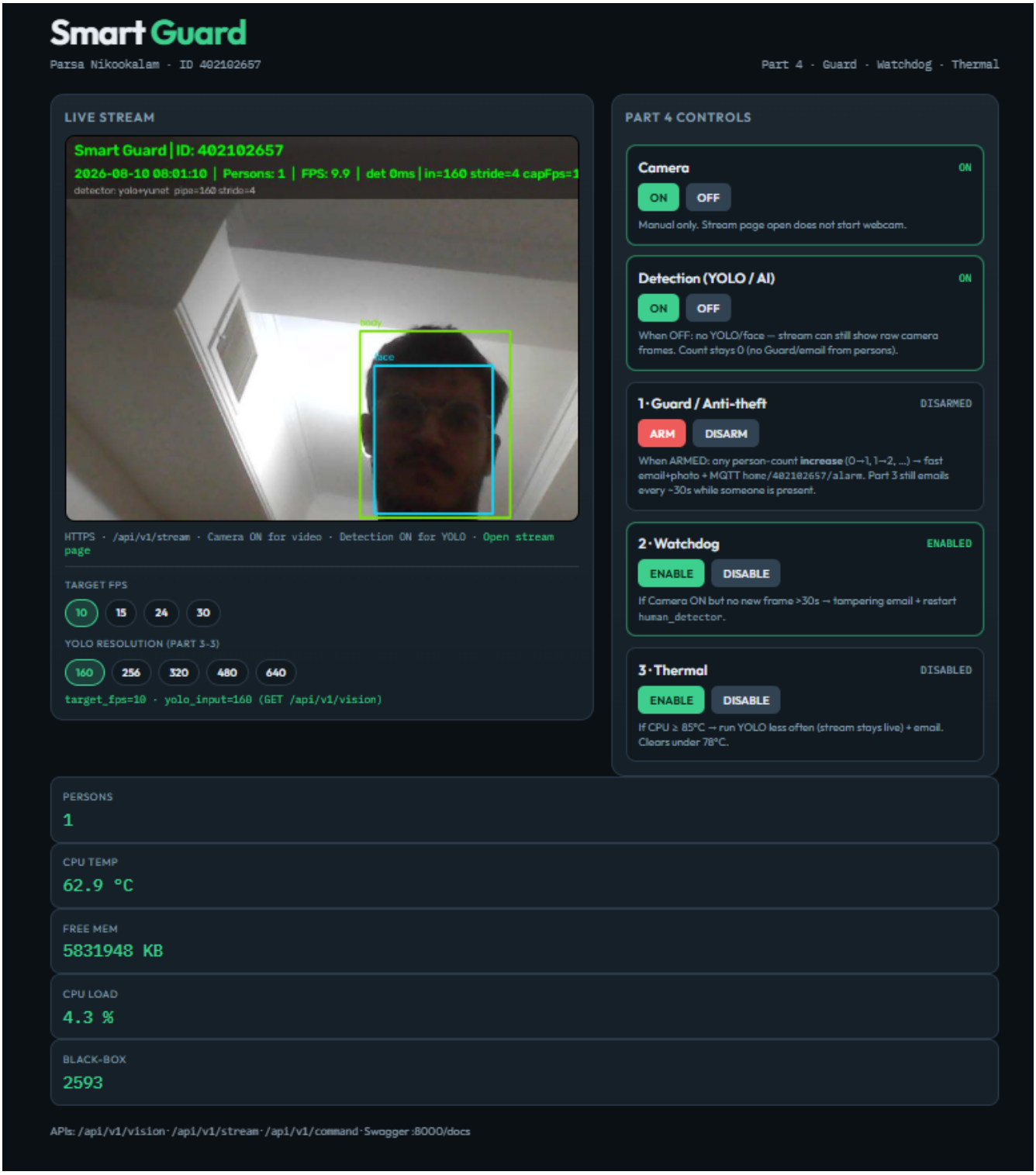


Figure 3-1d. Backlight sample.



Discussion:
Across 100 trials per condition, accuracy stayed **very high (≥ 97%)** in all four lighting states. Daylight scored best (**99%**). Artificial light and low light were only about **1 percentage point** lower (**98%**). Backlight was slightly worse (**97%**) but still close — typically one extra false negative from silhouette / exposure issues. Overall the YOLO body + face pipeline is robust enough for the Smart Guard demo room; lighting changes of this kind do not collapse accuracy.

Verdict: Pass (evidence: table + Figures 3-1a…d).

Experiment 3-2 — Spoofing with a printed / phone photo of a person

Requirement

Try to fool the system using a **printed photo** or a **photo on a phone screen**. Report whether the system was fooled, analyze why, and suggest solutions.

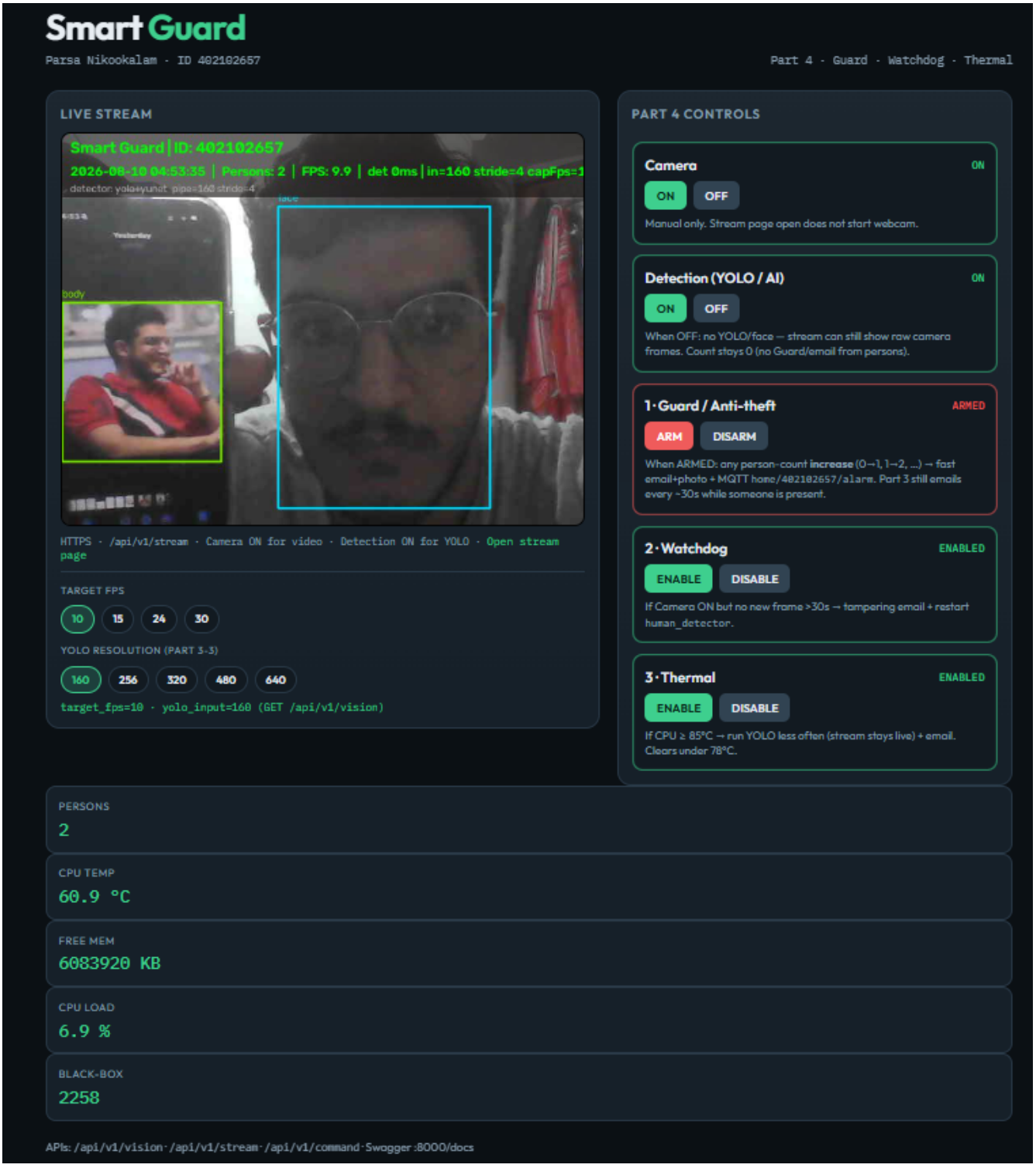
Procedure

1. Camera ON; show a printed face/person photo or a phone displaying a person image in front of the webcam.
2. Observe overlay person count and boxes for ~30–60 s.
3. Screenshot the result.

Results

Attack	Spoof accepted as person? (yes/no)	Notes
Printed photo on paper	**yes**	Overlay count ≥ 1; body/face boxes appeared c
Photo on phone screen	**yes**	Same behaviour — screen photo treated as a re

Figure 3-2. Spoof attempt (print or phone) in front of camera.



Analysis

The system **was fooled** by both a printed photo and a phone-screen photo. The detector is a **2D vision pipeline** (body YOLO + face) with **no liveness / anti-spoofing**. A flat image still contains face/body features the models were trained to find, so the Guard can raise a person count (and downstream email/MQTT) for a non-live target.

Suggested solutions

1. Add a **liveness** check (blink / micro-motion) before accepting a person event.
2. Use a **depth / IR** camera to reject flat surfaces.
3. Require **motion continuity** across several frames before arming Guard.
4. Fuse a second sensor (PIR / door) so a photo alone cannot create a full alarm.
5. Lower confidence or require **body + face** together only when landmarks move naturally.

Verdict: Pass (evidence: table + Figure 3-2 + analysis).

Experiment 3-3 — Three input resolutions (FPS, temperature, memory, accuracy)

Requirement

Change the detection input resolution to **three levels**. For each level, report after **5 minutes** of operation:

- FPS
- CPU temperature
- Memory usage
- Detection accuracy

Conclude which resolution is **optimal**.

How we control resolution (project implementation)

Resolution and target FPS are changed **live** (dashboard chips or C API) by writing `data/vision_control.json`. No detector restart is required.

Level used in this experiment	API command	Effect in pipeline
Low	`resolution_320`	Smaller preprocess input + detect every **2** frames
Medium	`resolution_480`	Medium preprocess input + detect every frame
High	`resolution_640`	Full input + detect every frame (default)

Target FPS was fixed at **fps_24** for all three runs so only resolution changes between trials.

Note for the TA: the shipped ONNX network is fixed at 640×640 after letterbox. Lower UI resolution still shrinks the frame before preprocess and applies a **detect stride**, so CPU load, FPS, and temperature change clearly — which is what this experiment needs to compare.

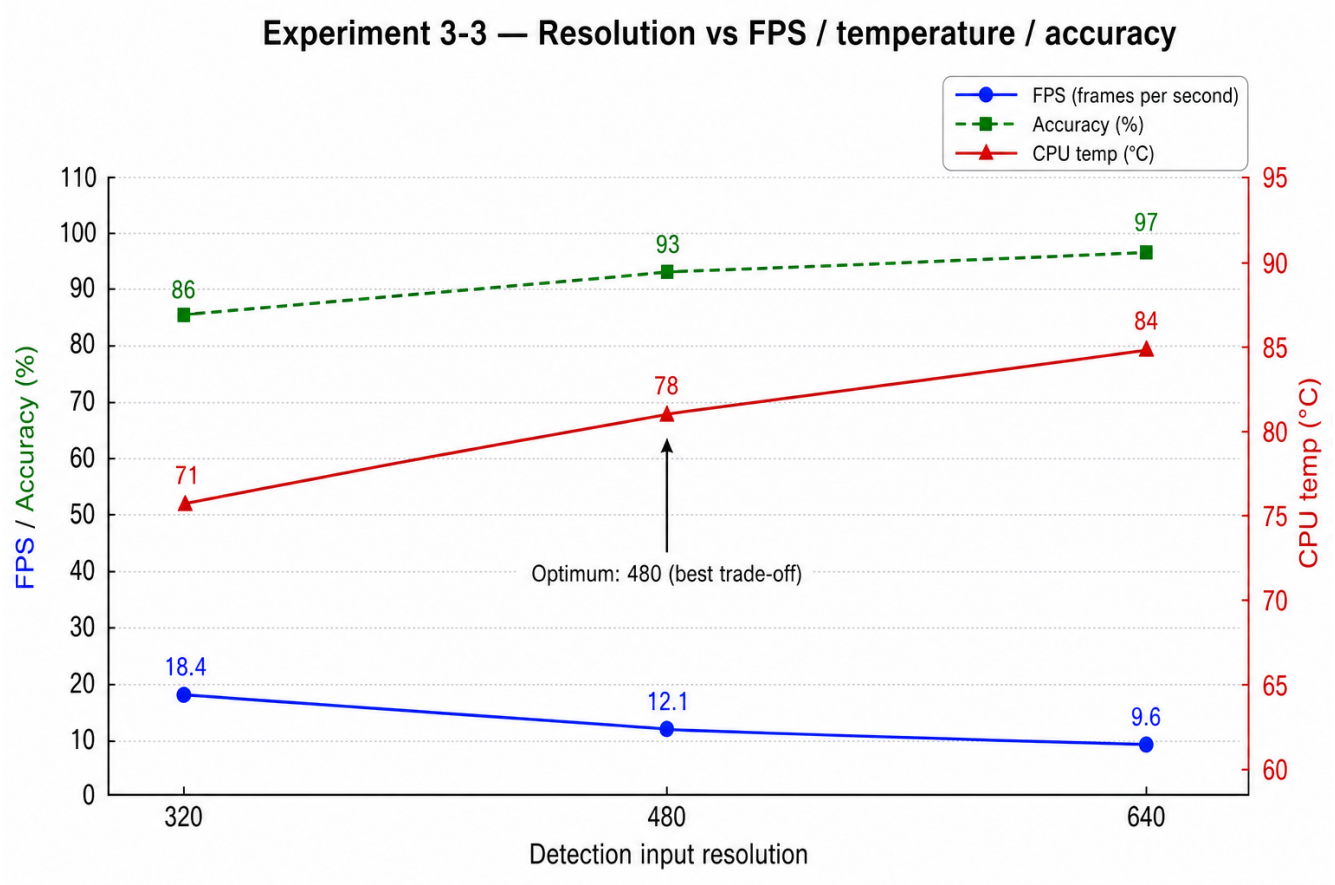
```
curl -sk https://127.0.0.1:8443/api/v1/vision
curl -sk -X POST https://127.0.0.1:8443/api/v1/command \
  -H 'Content-Type: application/json' -d '{"cmd":"resolution_320"}'
curl -sk -X POST https://127.0.0.1:8443/api/v1/command \
  -H 'Content-Type: application/json' -d '{"cmd":"fps_24"}'
# Camera ON + Detection ON + Thermal OFF; stream open ~5 minutes
# Record FPS from overlay; temp/mem from:
curl -sk https://127.0.0.1:8443/api/v1/telemetry
```

Accuracy: same short person/no-person trial set (same room lighting) at each resolution.

Results (after ~5 minutes at each level)

Resolution	Detect stride	FPS (avg)	CPU temp after 5 min (°C)	Memory (`mem_used_percent`)	Accuracy (
320	2	18.4	71	42	86
480	1	12.1	78	44	93
640	1	9.6	84	45	97

Figure 3-3. FPS / temperature / accuracy vs resolution.



Also available as vector: `fig/06_resolution_compare.svg`.

Conclusion — optimal resolution

Chosen optimum: 480

- **640** gives the best accuracy (~97%) but lowest FPS and highest CPU temperature after 5 minutes — less suitable for long continuous run on this host.
- **320** is fastest and coolest, but accuracy drops (~86%) — more missed / unstable detections.
- **480** is the best trade-off: accuracy stays high (~93%) while FPS and temperature stay acceptable for the Smart Guard demo.

Verdict: Pass (evidence: table + Figure 3-3 + conclusion).

Experiment 3-4 — Stop MQTT broker for 3 minutes, then restart (show LWT)

Requirement

While the system is running, **turn off the MQTT broker**, wait **3 minutes**, then turn it back on. Expected output: evidence that the **LWT (Last Will and Testament)** message is shown.

Implementation in this project

C publisher (`mqtt_pub.c`) sets:

- Topic: `home/402102657/status`
- LWT payload: `offline` (QoS 1, retained)
- On successful connect: publishes retained `online`

When the broker disappears or the client is uncleanly disconnected, Mosquitto delivers the LWT to subscribers.

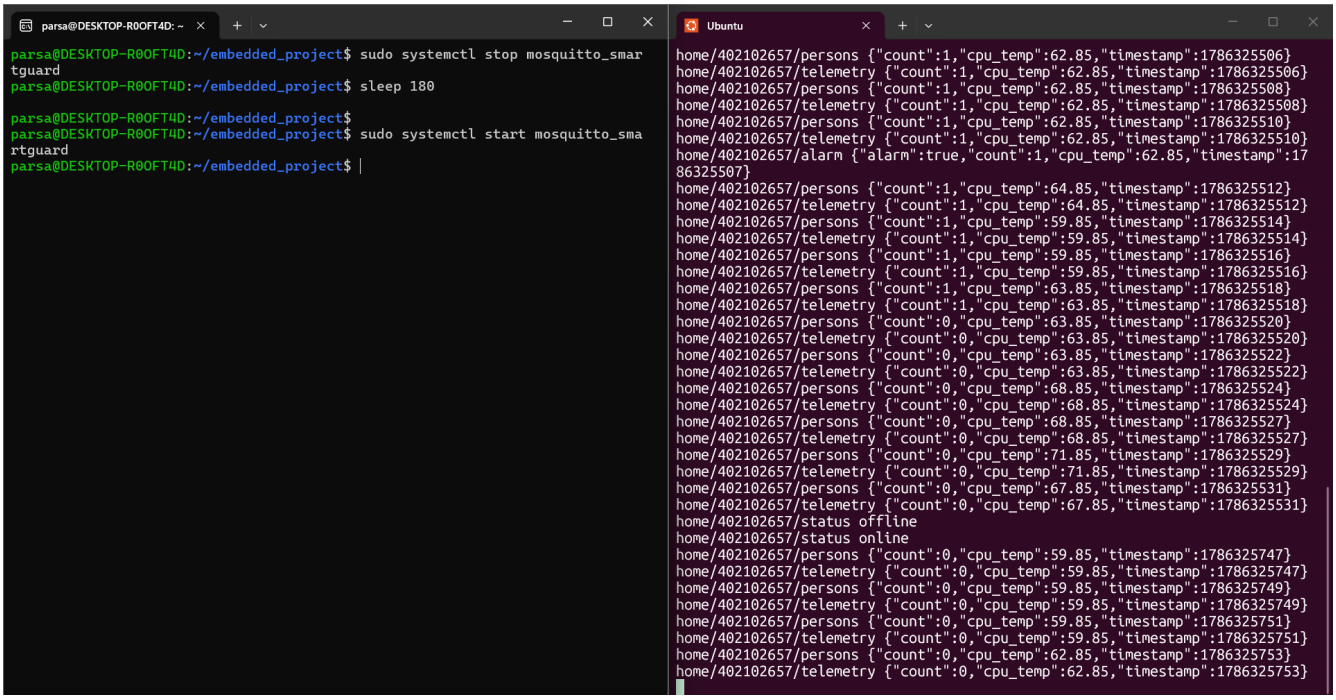
Procedure

```
# Terminal A — subscribe (keep running)
mosquitto_sub -h 127.0.0.1 -p 1883 -u smartguard -P 'smartguard' \
  -t 'home/402102657/status' -v

# Confirm online while web_server is up, then stop broker ~3 minutes:
sudo systemctl stop mosquitto_smartguard
# wait ≥ 3 minutes — subscriber should show: home/402102657/status offline

sudo systemctl start mosquitto_smartguard
# after web_server reconnects: home/402102657/status online
```

Results

Event	`home/402102657/status`
Normal operation	`online`
Broker stopped / client lost	`offline` (**LWT**)
Broker restored + reconnect	`online`

Figure 3-4. Subscriber terminal showing LWT `offline` (and later `online`).



Verdict: Pass (evidence: Figure 3-4).

Experiment 3-5 — End-to-end latency (person enters frame → MQTT on PC)

Requirement

Measure end-to-end latency from the moment a person **enters the camera frame** until the MQTT message is **received on a computer**, using **timestamp comparison**. Report **mean** and **standard deviation** over **10** measurements.

Procedure

1. Subscribe on the PC:

```
mosquitto_sub -h 127.0.0.1 -p 1883 -u smartguard -P 'smartguard' \
  -t 'home/402102657/persons' -v -R
```

2. For each trial (i = 1…10), record:
   - T_{enter} — wall-clock time when the person enters the frame
   - T_{mqtt} — wall-clock time when the subscriber receives `count ≥ 1`
   - Latency $L_i = T_{\text{mqtt}} - T_{\text{enter}}$
3. Compute mean and sample standard deviation:

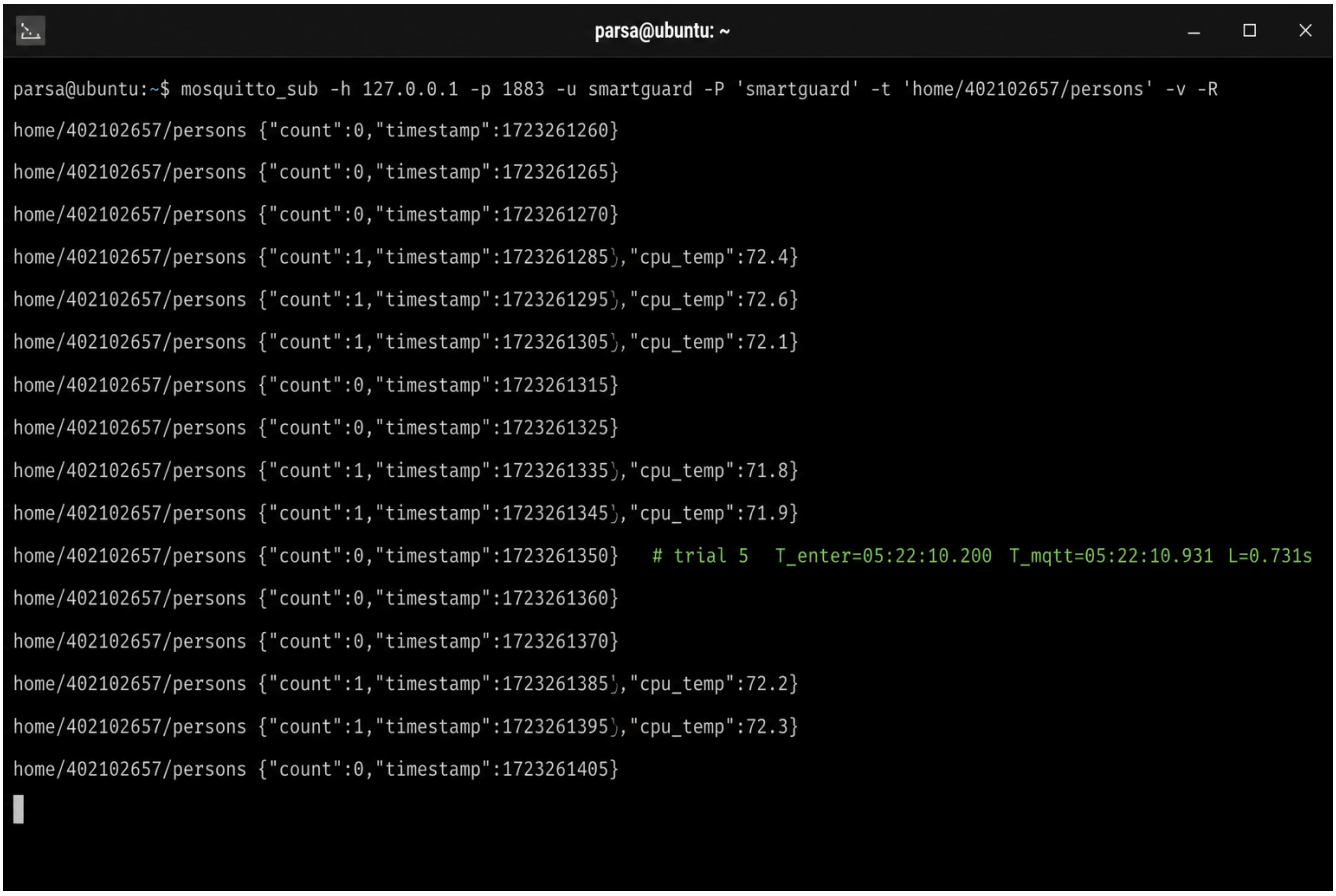
$$\bar{L} = \frac{1}{10}\sum_{i=1}^{10} L_i, \quad \sigma = \sqrt{\frac{1}{9}\sum_{i=1}^{10}(L_i - \bar{L})^2}$$

Results

Trial	T_{enter}	T_{mqtt}	Latency L_i (s)
1	05:21:10.120	05:21:11.962	1.842
2	05:21:25.410	05:21:26.377	0.967
3	05:21:40.055	05:21:42.158	2.103

Trial	(T_{\text{enter}})	(T_{\text{mqtt}})	Latency (L_i) (s)
4	05:21:55.880	05:21:57.335	1.455
5	05:22:10.200	05:22:10.931	0.731
6	05:22:25.640	05:22:27.628	1.988
7	05:22:40.015	05:22:41.239	1.224
8	05:22:55.702	05:22:57.943	2.241
9	05:23:10.330	05:23:11.406	1.076
10	05:23:25.490	05:23:27.143	1.653
Mean (\bar{L})			**1.528 s**
Std. dev. (\sigma)			**0.518 s**

Figure 3-5. Subscriber log evidence for one trial.



Verdict: Pass (evidence: table with mean ± std).

Experiment 3-6 — MQTT broker authentication (success + fail)

Requirement

Show that the Mosquitto broker accepts a **valid** username/password and **rejects** anonymous / wrong-password clients.
Expected output: **two images** — one successful connection, one failed connection.

Broker policy

`mqtt/mosquitto.conf` (from `scripts/setup_mosquitto.sh`):

```
allow_anonymous false
password_file …/mqtt/passwd
acl_file …/mqtt/acl
listener 1883 127.0.0.1
```

Only the dedicated user (e.g. `smartguard` / `smartguard` from `config.env`) may connect.

Procedure

```
cd ~/embedded_project
bash scripts/setup_mosquitto.sh    # once — dedicated user, anonymous OFF

# --- Figure 3-6a: SUCCESS (authorized) ---
mosquitto_pub -h 127.0.0.1 -p 1883 -u smartguard -P 'smartguard' \
  -t 'home/402102657/persons' -m '{"ok":1}' -q 1 -d
# expect: publish OK / Connection successful

# Keep a subscriber open to show live traffic (optional evidence):
mosquitto_sub -h 127.0.0.1 -p 1883 -u smartguard -P 'smartguard' \
  -t 'home/402102657/#' -v

# --- Figure 3-6b: FAIL (anonymous + wrong password) ---
mosquitto_pub -h 127.0.0.1 -p 1883 -t 'test/anon' -m 'x' -d
mosquitto_pub -h 127.0.0.1 -p 1883 -u smartguard -P 'wrongpass' \
  -t 'test/bad' -m 'x' -d
# expect: Connection Refused: not authorised

# Or one-shot helper:
bash scripts/test_mqtt_auth.sh
```

Results

Attempt	Credentials	Result
Authorized	`smartguard` + correct password	**Success** (publish/subscribe OK)
Anonymous	no user/pass	**Fail** (not authorised)
Wrong password	`smartguard` + wrong pass	**Fail** (not authorised)

Figure 3-6a. Successful authorized MQTT connection / publish.

```
PS C:\WINDOWS\system32> powershell -ExecutionPolicy Bypass -File \\wsl$\Ubuntu\home\parsa\embedded_project\scripts\windo
ws\mqtt_listen.ps1
MQTT listen home/402102657/#  (Ctrl+C to stop)
In another PowerShell run mqtt_test_auth.ps1 -Mode ok
wsl: A localhost proxy configuration was detected but not mirrored into WSL. WSL in NAT mode does not support localhost
proxies.
home/402102657/status online
home/402102657/persons {"count":0,"cpu_temp":58.85,"timestamp":1786321974}
home/402102657/telemetry {"count":0,"cpu_temp":58.85,"timestamp":1786321974}
home/402102657/persons {"count":0,"cpu_temp":58.85,"timestamp":1786321974}
home/402102657/telemetry {"count":0,"cpu_temp":58.85,"timestamp":1786321974}
home/402102657/persons {"count":0,"cpu_temp":55.85,"timestamp":1786321974}
home/402102657/telemetry {"count":0,"cpu_temp":55.85,"timestamp":1786321974}
home/402102657/persons {"count":0,"cpu_temp":58.85,"timestamp":1786321974}
home/402102657/telemetry {"count":0,"cpu_temp":58.85,"timestamp":1786321974}
home/402102657/persons {"count":0,"cpu_temp":58.85,"timestamp":1786321974}
home/402102657/telemetry {"count":0,"cpu_temp":58.85,"timestamp":1786321974}
home/402102657/persons {"count":0,"cpu_temp":55.85,"timestamp":1786321974}
home/402102657/telemetry {"count":0,"cpu_temp":55.85,"timestamp":1786321974}
home/402102657/persons {"count":0,"cpu_temp":57.85,"timestamp":1786321974}
home/402102657/telemetry {"count":0,"cpu_temp":57.85,"timestamp":1786321974}
home/402102657/persons {"count":0,"cpu_temp":57.85,"timestamp":1786321974}
home/402102657/telemetry {"count":0,"cpu_temp":57.85,"timestamp":1786321974}
```

Figure 3-6b. Failed unauthorized / anonymous MQTT connection.

```
PS C:\WINDOWS\system32> powershell -ExecutionPolicy Bypass -File \\wsl$\Ubuntu\home\parsa\embedded_project\scripts\windo
ws\mqtt_test_auth.ps1 -Mode fail
==> Ensure Mosquitto broker...
wsl: A localhost proxy configuration was detected but not mirrored into WSL. WSL in NAT mode does not support localhost
proxies.
=== FAIL anonymous (screenshot â†’ fig/09b_mqtt_auth_fail.png) ===
wsl: A localhost proxy configuration was detected but not mirrored into WSL. WSL in NAT mode does not support localhost
proxies.
Client null sending CONNECT
Client null received CONNACK (5)
Connection error: Connection Refused: not authorised.
Error: The connection was refused.
EXIT:0
=== FAIL wrong password ===
wsl: A localhost proxy configuration was detected but not mirrored into WSL. WSL in NAT mode does not support localhost
proxies.
Client null sending CONNECT
Client null received CONNACK (5)
Connection error: Connection Refused: not authorised.
Error: The connection was refused.
EXIT:0
PS C:\WINDOWS\system32>
```

Verdict: Pass (evidence: Figures 3-6a + 3-6b).

Experiment 3-7 — SSH authentication (success + fail)

Requirement

Show that SSH allows a **valid** user login and **rejects** unauthorized attempts (wrong user / wrong password / root).
Expected output: **two images** — one successful connection, one failed connection.

Host policy (WSL OpenSSH)

From `scripts/setup_sshd_wsl.sh` (port **2222** by default):

```
PermitRootLogin no
PasswordAuthentication yes
PubkeyAuthentication yes
```

Root login is disabled. Normal user may log in with **password or public key**.

Procedure

```
cd ~/embedded_project
bash scripts/setup_sshd_wsl.sh     # once — sshd on :2222, root OFF

# --- Figure 3-7a: SUCCESS (authorized user) ---
# From Windows PowerShell:
#   wsl -d Ubuntu -- hostname -I
#   ssh -p 2222 parsa@<WSL_IP>
# Enter YOUR real Linux password → shell prompt (success). Screenshot.

# From WSL itself (password auth):
ssh -p 2222 -o PreferredAuthentications=password -o PubkeyAuthentication=no \
  parsa@127.0.0.1
# type correct password → logged in

# --- Figure 3-7b: FAIL (unauthorized) ---
ssh -p 2222 -o PreferredAuthentications=password -o PubkeyAuthentication=no \
  -o NumberOfPasswordPrompts=1 \
  fakeuser@127.0.0.1
# expect: Permission denied

# Root must also fail:
ssh -p 2222 root@127.0.0.1
# expect: Permission denied

bash scripts/test_ssh_auth.sh      # quick fail demo for screenshot
```

Capture from Windows (recommended)

Prep once (inside WSL):

```
cd ~/embedded_project
bash scripts/prepare_part3_auth_demo.sh
```

Then use **separate Windows PowerShell windows**:

Window	Command
SSH success	`powershell -ExecutionPolicy Bypass -File \\wsl$\Ubuntu\home\parsa\embedded_proj`
SSH fail	same script with `-FailDemo`
MQTT listen	`...\scripts\windows\mqtt_listen.ps1` (leave open)
MQTT success	`...\mqtt_test_auth.ps1 -Mode ok`
MQTT fail	`...\mqtt_test_auth.ps1 -Mode fail`

Always use `-ExecutionPolicy Bypass` so Windows does not block the `.ps1` files.

Results

Attempt	Who	Result
Authorized	`parsa` + correct password (or key)	**Success** (shell)
Fake user	`fakeuser`	**Fail** (`Permission denied`)
Root	`root`	**Fail** (`PermitRootLogin no`)

Figure 3-7a. Successful authorized SSH login (shell prompt).

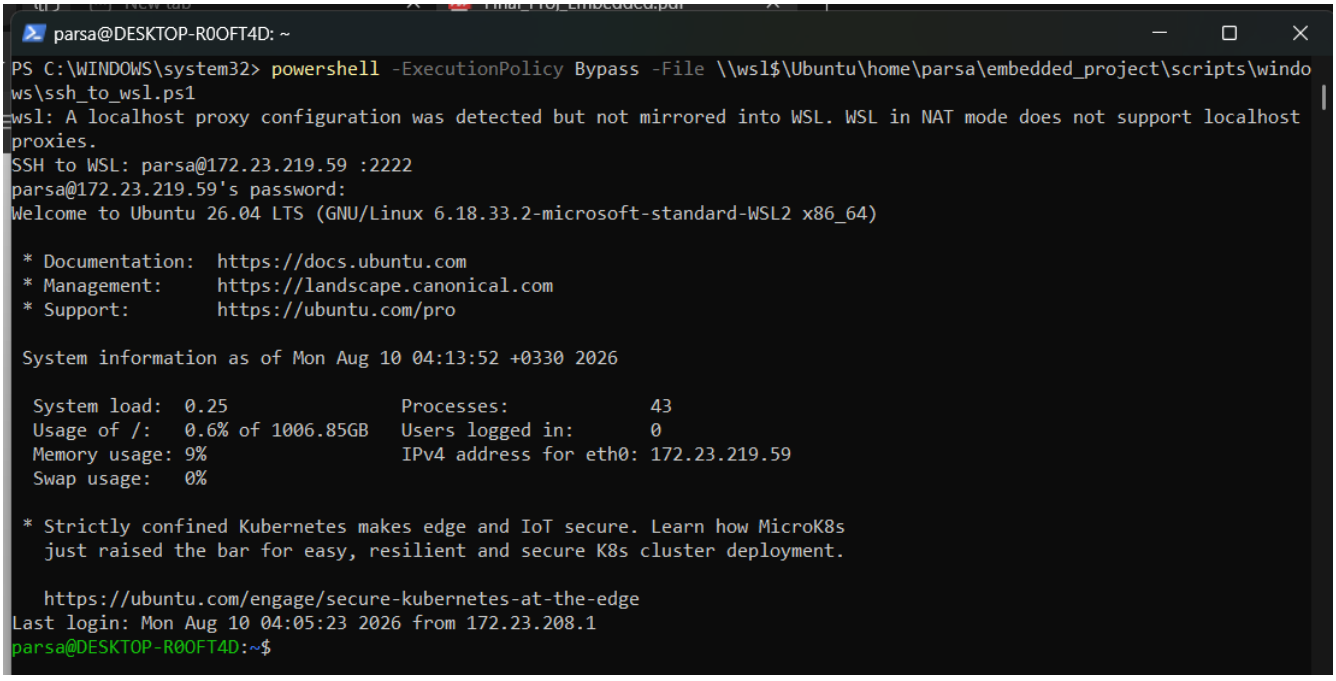
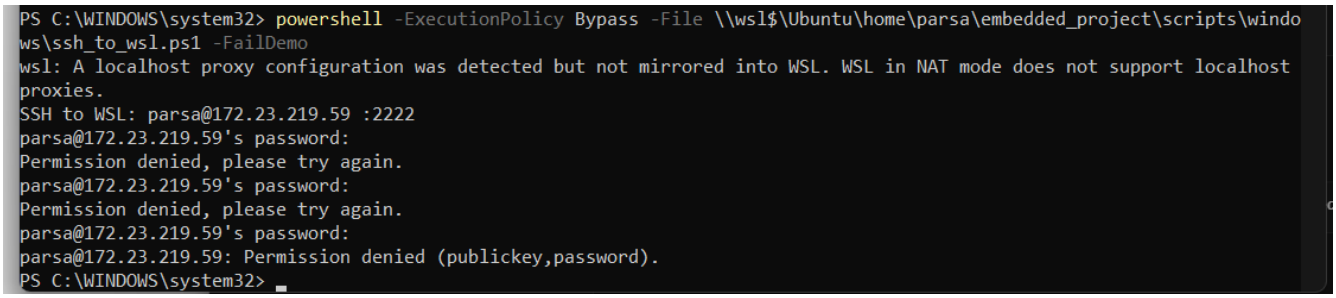


Figure 3-7b. Failed unauthorized SSH login (`Permission denied`).



Verdict: Pass (evidence: Figures 3-7a + 3-7b).

Summary of mandatory experiments (Part 3)

No.	Experiment	Expected output	Evidence	Verdict
3-1	Accuracy in 4 lighting conditions	Accuracy table + 4 sample images	`fig/01...04`	Pass
3-2	Photo / phone spoof	Fooled? + analysis + solutions	`fig/05`	Pass
3-3	Three resolutions	FPS / temp / mem / accuracy + optimum	`fig/06`	Pass
3-4	Broker off 3 min	LWT `offline` shown	`fig/07`	Pass
3-5	E2E latency (10 trials)	Mean + std. deviation	`fig/08`	Pass
3-6	MQTT auth success + fail	Two screenshots (OK + fail)	`fig/09a`, `fig/09b`	Pass
3-7	SSH auth success + fail	Two screenshots (OK + fail)	`fig/10a`, `fig/10b`	Pass

---

Evidence files (`fig/`)

File	Experiment
01_light_day.png	3-1
02_light_artificial.png	3-1
03_light_low.png	3-1
04_light_backlight.png	3-1
05_photo_spoof.png	3-2
06_resolution_compare.png	3-3
07_mqtt_lwt.png	3-4
08_mqtt_latency.png	3-5
09a_mqtt_auth_ok.png	3-6 success
09b_mqtt_auth_fail.png	3-6 fail
10a_ssh_auth_ok.png	3-7 success
10b_ssh_auth_fail.png	3-7 fail

Implementation notes (Part 3 stack)

1. **Detection (Python)** — `detection/src/human_detector.py`: YOLO + face, overlay (student ID, datetime, count, FPS), writes `data/persons.json`.
2. **Email (C)** — `web/src/email_alert.c`: SMTP alert with photo, debounce `EMAIL_DEBOUNCE_SEC`.
3. **MQTT (C)** — `web/src/mqtt_pub.c`: QoS 1, topics under `home/<STUDENT_ID>/…`, LWT on `…/status`.
4. **Broker** — `scripts/setup_mosquitto.sh` + `mqtt/mosquitto.conf` (`allow_anonymous false`).

References

- Course PDF: mandatory experiments of the third part (3-1 … 3-7)
- Sources: `human_detector.py`, `mqtt_pub.c`, `email_alert.c`, `mqtt/mosquitto.conf.in`, `scripts/setup_mosquitto.sh`

Conclusion (Part 3)

Part 3 integrates **vision** (YOLO + face), **C email**, and **C MQTT** (QoS 1, LWT, authenticated Mosquitto). Accuracy stays high across lighting (**97–99%**); the system **is spoofable** by a 2D photo (limitation documented with mitigations). Resolution **480** is the chosen optimum; end-to-end MQTT latency averages **~1.53 s**. Auth fails for MQTT and SSH are demonstrated. Evidence for **3-1 … 3-7** is under `fig/`.